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Unit 08: Data Preprocessing Using Rapid Miner

CONTENTS

Objectives

After this lecture, you will be able to

- Learn the need for preprocessing of data.
- Need and implementation of Remove Duplicate Operator.
- Learn the implementation of Different methods of Handing Missing Values.
- Understand the various methods of renaming an attribute.
- Learn the implementation of the Apriori Algorithm in Weka.

Introduction

Data pre-processing is a data mining technique that entails converting raw data into a format that can be understood. Real-world data is often incomplete, unreliable, and/or deficient in specific habits or patterns, as well as containing numerous errors. The most important information at the pre-processing phase is about the missing values.Preprocessing data is a data mining technique for transforming raw data into a usable and efficient format. The measures taken to make data more suitable for data mining are referred to as data preprocessing. The steps involved in Data Preprocessing are normally divided into two categories:

- Selecting data objects and attributes for analysis, and selecting data objects and attributes for analysis.
- Adding/removing attributes is a two-step process.

8.1 Remove Duplicate Operator

The Remove Duplicates operator compares all examples in an Example Set against each other using the listed attributes to remove duplicates. This operator eliminates duplicate examples one by one, leaving only one duplicate example. If the selected attributes have the same values, two instances are called duplicates.

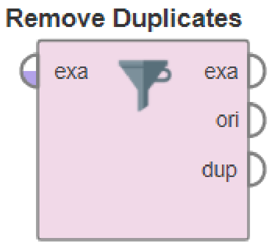


Figure 1: Remove Duplicate Operator

The attribute filter type parameter and other related parameters can be used to select attributes. Assume that two attributes, 'att1' and 'att2,' are chosen, with three and two possible values, respectively. As a result, there are a total of six (three times two) special combinations of these two attributes. As a consequence, the resulting Example Set can only have 6 examples. This operator applies to all attribute types. The following Iris dataset contains 150 examples before the use of remove duplicate operator. The following are the various parameters applicable to remove duplicate operators.

Consider any dataset from sample dataset in Rapid miner and implement Remove Duplicate operator on different attributes.

Input

An Example Set is expected at this input port. In the attached Example Method, it's the result of the Retrieve operator. Other operators' output can also be used as data.

Output

- Example set Output(Data Table)

The duplicate examples in the provided Example Set are removed, and the resulting Example Set is delivered via this port.

- Original (Data Table)

This port passes the Example Set that was provided as input to the output without any changes. This is commonly used to reuse the same Example Set through several operators or to display the Example Set in the Results Workspace.

- Duplicates (Data Table)

This port is used to deliver duplicated examples from the given Example Set.

Parameters

attribute_filter_type

This parameter lets you choose the attribute selection filter, which is the tool you'll use to pick the appropriate attributes. It comes with the following features:

- **All**: This choice simply selects all of the Example Set's attributes. This is the standard-setting.
- **single:** Selecting a single attribute is possible with this choice. When you select this choice, a new parameter (attribute) appears in the Parameters panel.
- **subset:** This choice allows you to choose from a list of multiple attributes. The list contains all of the Example Set's attributes; appropriate attributes can be easily selected. If the metadata is unknown, this option will not work.
- **regular_expression:** This feature allows you to pick attributes using a standard expression. Other parameters (regular expression, use except for expression) become available in the Parameters panel when this choice is selected.

- **value_type:** This option allows you to pick all of a type's attributes. It's worth noting that styles are arranged in a hierarchy. Real and integer types, for example, are also numeric types. When selecting attributes via this option, users should have a basic understanding of type hierarchy. Other parameters in the Parameters panel become available when this choice is selected.
- **block_type:** This option works similarly to the value type option. This choice allows you to choose all of the attributes for a specific block form. Other parameters (block type, use block type exception) become available in the Parameters panel when this choice is selected.
- **no_missing_values**: This choice simply selects all of the Example Set's attributes that do not have a missing value in any of the examples. All attributes with a single missing value are deleted.
- **numeric value filter:** When the numeric value filter choice is selected, a new parameter (numeric condition) appears in the Parameters panel. The examples of all numeric attributes that satisfy the numeric condition are chosen. Please note that regardless of the numerical situation, all nominal attributes are chosen.

attribute

This choice allows you to choose the desired attribute. If the metadata is identified, the attribute name can be selected from the attribute parameter's drop-down box.

attributes

This choice allows you to pick the appropriate attributes. This will bring up a new window with two lists. The left list contains all attributes, which can be transferred to the right list, which contains the list of selected attributes for which the conversion from nominal to numeric will be performed; all other attributes will remain unchanged.

Regular_expression

This expression will be used to pick the attributes whose names match this expression. Regular expressions are a powerful tool, but they require a thorough introduction for beginners. It's always a good idea to use the edit and display regular expression menu to define the regular expression.

value_type

A drop-down menu allows you to pick the type of attribute you want to use. You may choose one of the following types: nominal, text, binominal, polynomial, or file path.

The following figure shows the design view of the Iris dataset.

block_type

A drop-down list may be used to choose the block type of attributes to be used. 'single value' is the only possible value.

numeric_condition

This is where you specify the numeric condition for checking examples of numeric attributes. The numeric condition '> 6', for example, will hold all nominal and numeric attributes with a value greater than 6 in any example. '> 6 && 11' or '= 5 || 0' are examples of potential situations. However, you can't use && and || in the same numeric situation.

Did you know? Conditions like '(> 0 && 2) || (>10 && 12)' are not authorized, since they use both && and ||. After '>', '=', and '', use a blank space; for example, '5' will not work; instead, use '5'.

include_special_attributes

The examples are identified by unique attributes, which are attributes with specific functions. Regular attributes, on the other hand, simply define the instances. Id, label, prediction, cluster, weight, and batch are special attributes.

invert_selection

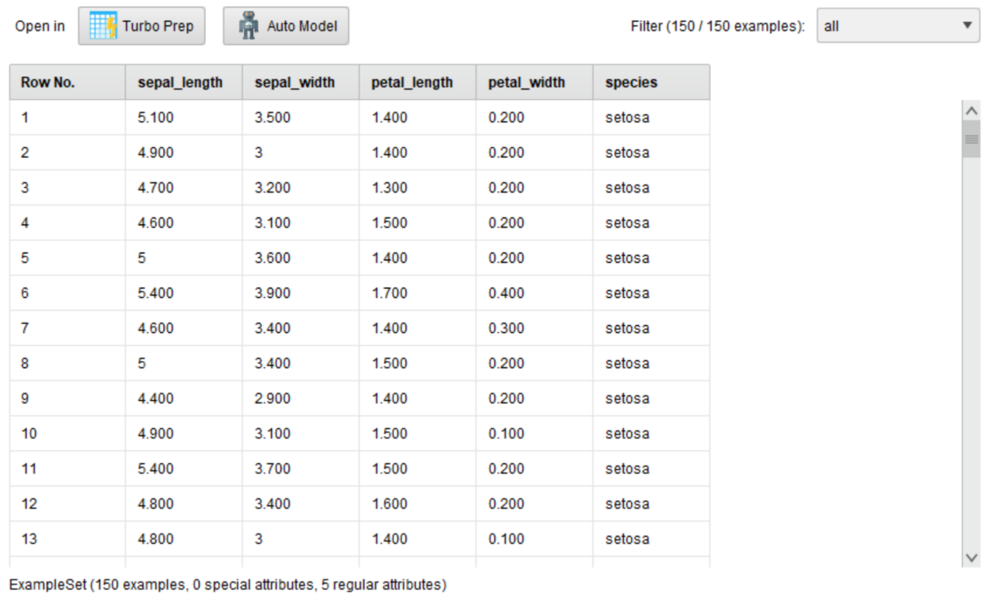
When set to real, this parameter acts as a NOT gate, reversing the selection. In that case, all of the previously selected attributes are unselected, while previously unselected attributes are selected. Before checking this parameter, for example, if attribute 'att1' is selected and attribute 'att2' is unselected. 'att1' will be unselected after testing this parameter, while 'att2' will be selected.

treat_missing_values_as_duplicates

This parameter determines whether or not missing values should be considered duplicates. If valid, missing values are treated the same as duplicate values.

Figure 2:Design View

Figure 3 shows the result view of the Iris dataset.



Open in Turbo Prep Auto Model Filter (150 / 150 examples): all

Row No.	sepal_length	sepal_width	petal_length	petal_width	species
1	5.100	3.500	1.400	0.200	setosa
2	4.900	3	1.400	0.200	setosa
3	4.700	3.200	1.300	0.200	setosa
4	4.600	3.100	1.500	0.200	setosa
5	5	3.600	1.400	0.200	setosa
6	5.400	3.900	1.700	0.400	setosa
7	4.600	3.400	1.400	0.300	setosa
8	5	3.400	1.500	0.200	setosa
9	4.400	2.900	1.400	0.200	setosa
10	4.900	3.100	1.500	0.100	setosa
11	5.400	3.700	1.500	0.200	setosa
12	4.800	3.400	1.600	0.200	setosa
13	4.800	3	1.400	0.100	setosa

ExampleSet (150 examples, 0 special attributes, 5 regular attributes)

Figure 3: Iris Dataset before removal of Duplicates

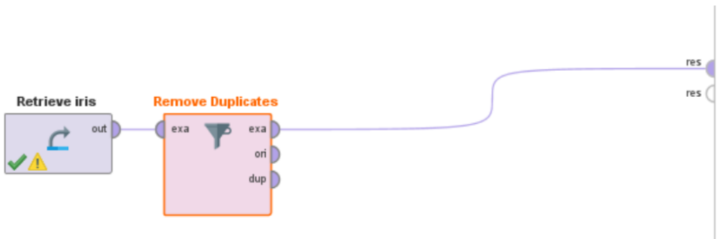
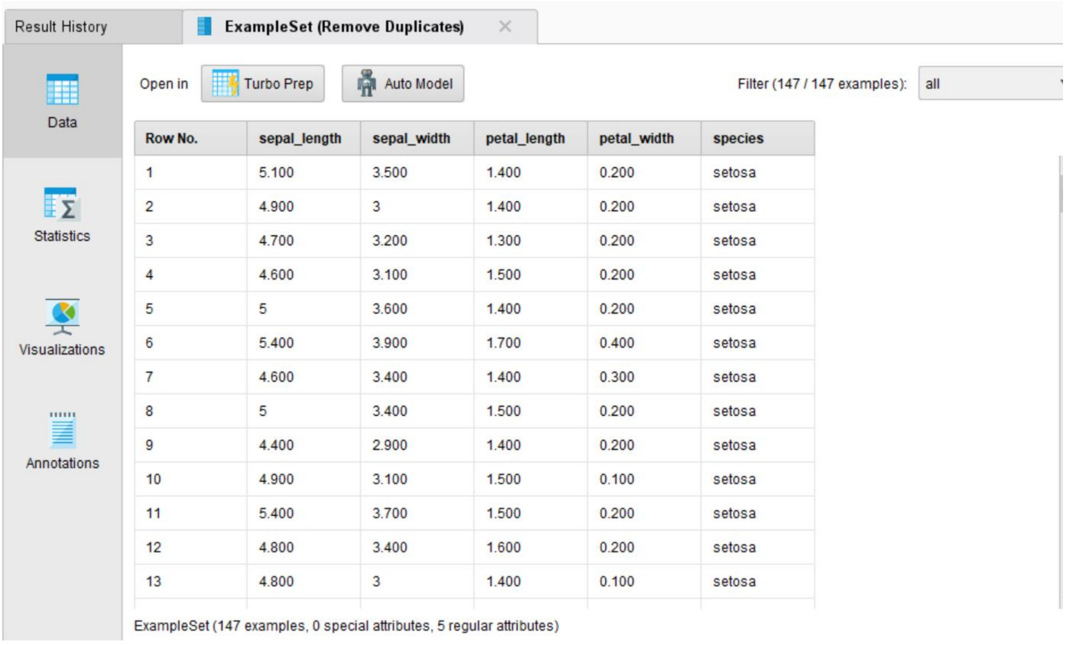


Figure 4: Design View using Remove Duplicate Operator



Row No.	sepal_length	sepal_width	petal_length	petal_width	species
1	5.100	3.500	1.400	0.200	setosa
2	4.900	3	1.400	0.200	setosa
3	4.700	3.200	1.300	0.200	setosa
4	4.600	3.100	1.500	0.200	setosa
5	5	3.600	1.400	0.200	setosa
6	5.400	3.900	1.700	0.400	setosa
7	4.600	3.400	1.400	0.300	setosa
8	5	3.400	1.500	0.200	setosa
9	4.400	2.900	1.400	0.200	setosa
10	4.900	3.100	1.500	0.100	setosa
11	5.400	3.700	1.500	0.200	setosa
12	4.800	3.400	1.600	0.200	setosa
13	4.800	3	1.400	0.100	setosa

Figure 5: Iris dataset Removing Duplicates

Originally there is a total of 150 examples in the Iris dataset after the implementation of the remove duplicate operator we have left with 147 examples rest 3 examples were duplicates which are removed with the remove duplicate operator.

If the selected attributes have the same values, two instances are called duplicate. The attribute filter type parameter and other related parameters can be used to select attributes. Assume that two attributes, 'att1' and 'att2,' are chosen, with three and two possible values, respectively.

8.2 Rename an Attribute

The following are the various operators used for the renaming task.

Rename

The Rename operator is used to rename one or more of the input ExampleSet's attributes. It's important to remember that attribute names must be special. The Rename operator does not affect an attribute's form or position.

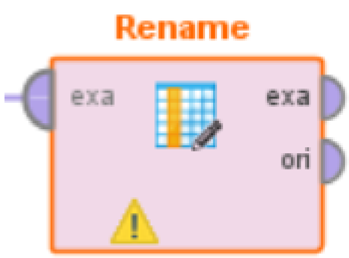


Figure 6: Rename Operator

For example, suppose you have an integer form and standard role attribute called 'alpha.' The attribute's name would be changed if it is renamed to 'beta.' It will keep its integer form and daily function. Use the Set Function operator to adjust an operator's role. At 'Data Transformation/Class Conversion,' a variety of type conversion operators are available for modifying the type of an attribute.

The Rename operator takes two parameters; one is the existing attribute name and the other is the new name you want the attribute to be renamed to.

The Rename operator has the following ports:

Input

An Example Set is expected at this input port. In the attached Example Method, it's the result of the Retrieve operator. Other operators' output can also be used as data. Since attributes are defined in its metadata, metadata must be attached to data for input. The Retrieve operator returns metadata in addition to data.

Output

This port produces an Example Set with renamed attributes.

Original

This port passes the Example Set that was provided as input to the output without any changes. This is commonly used to reuse the same Example Set through several operators or to display the Example Set in the Results Workspace.

Parameters

old name

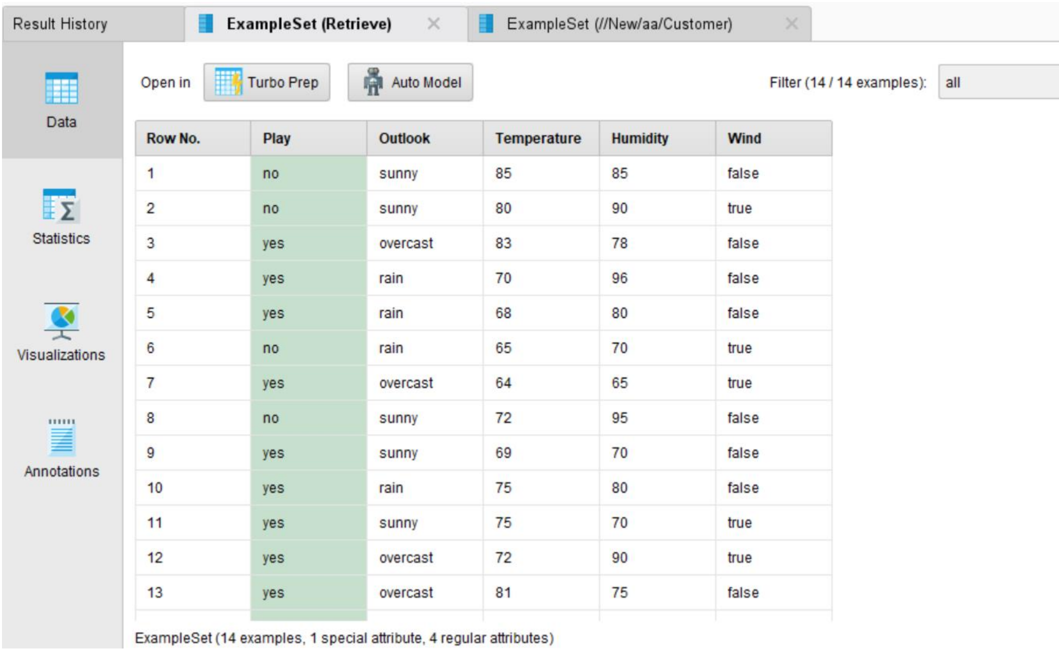
This parameter is used to specify which attribute's name should be modified.

new name

This parameter is used to specify the attribute's new name. Special characters may also be used in a name.

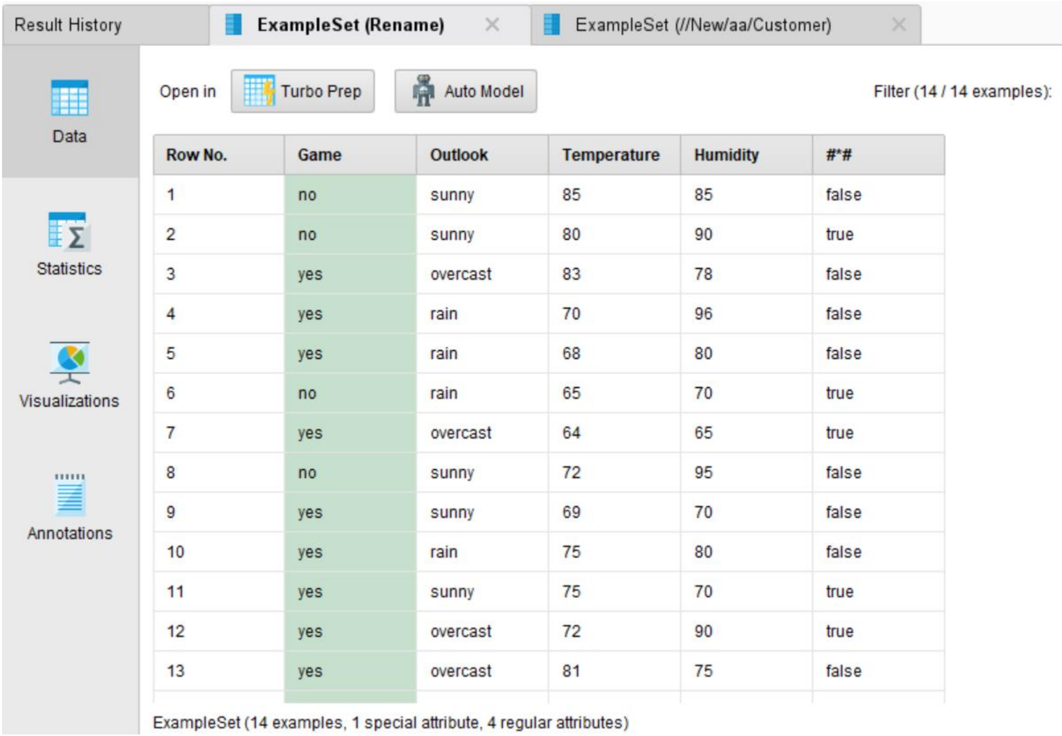
rename additional attributes

Click the Edit List button to rename several attributes. You can choose attributes and give them new names here.



Row No.	Play	Outlook	Temperature	Humidity	Wind
1	no	sunny	85	85	false
2	no	sunny	80	90	true
3	yes	overcast	83	78	false
4	yes	rain	70	96	false
5	yes	rain	68	80	false
6	no	rain	65	70	true
7	yes	overcast	64	65	true
8	no	sunny	72	95	false
9	yes	sunny	69	70	false
10	yes	rain	75	80	false
11	yes	sunny	75	70	true
12	yes	overcast	72	90	true
13	yes	overcast	81	75	false

Figure 7: Exampleset before Rename Operator



Result History | ExampleSet (Rename) | ExampleSet (//New/aa/Customer)

Data | Statistics | Visualizations | Annotations

Open in | Turbo Prep | Auto Model | Filter (14 / 14 examples):

Row No.	Game	Outlook	Temperature	Humidity	#*#
1	no	sunny	85	85	false
2	no	sunny	80	90	true
3	yes	overcast	83	78	false
4	yes	rain	70	96	false
5	yes	rain	68	80	false
6	no	rain	65	70	true
7	yes	overcast	64	65	true
8	no	sunny	72	95	false
9	yes	sunny	69	70	false
10	yes	rain	75	80	false
11	yes	sunny	75	70	true
12	yes	overcast	72	90	true
13	yes	overcast	81	75	false

ExampleSet (14 examples, 1 special attribute, 4 regular attributes)

Figure 8:Resulset After Implementing Rename Operator

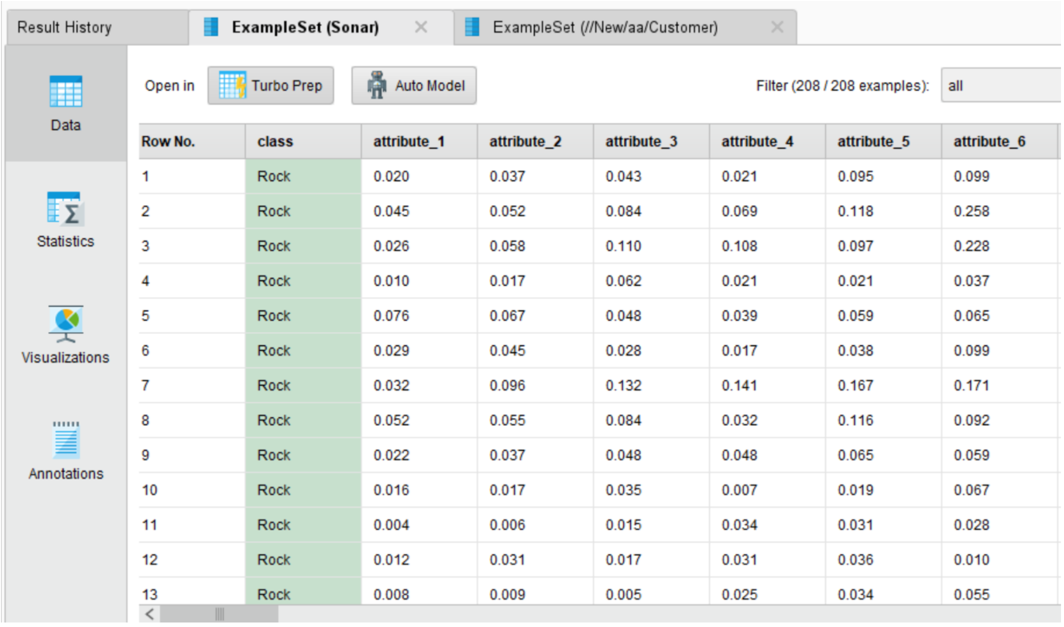
This Example Process makes use of the 'Golf' data collection. The 'Wind' attribute has been renamed to '#*#', and the 'Play' attribute has been renamed to 'Game'. To demonstrate that special characters can be used to rename attributes, the 'Wind' attribute is renamed to '#*#'. Attribute names, on the other hand, should always be meaningful and specific to the type of data stored in them.

Rename By replacing

This operator can be used to rename a collection of attributes by substituting a substitute for parts of the attribute names.

Figure 9: Rename by Replacing Operator

The Rename by Replacing operator substitutes the required replacement for parts of the attribute names. This operator is often used to exclude unnecessary characters from attribute names, such as whitespaces, parentheses, and other characters. The substitute parameter specifies which part of the attribute name should be changed. It's a normal term, and it's a very strong one at that.An arbitrary string may be used as the replace by parameter. Empty strings are also permitted. Using $1, $2, $3, and so on to capture groups of the regular expression of the replace what parameter can be accessed. Please take a look at the attached Example Process for more details.



Row No.	class	attribute_1	attribute_2	attribute_3	attribute_4	attribute_5	attribute_6
1	Rock	0.020	0.037	0.043	0.021	0.095	0.099
2	Rock	0.045	0.052	0.084	0.069	0.118	0.258
3	Rock	0.026	0.058	0.110	0.108	0.097	0.228
4	Rock	0.010	0.017	0.062	0.021	0.021	0.037
5	Rock	0.076	0.067	0.048	0.039	0.059	0.065
6	Rock	0.029	0.045	0.028	0.017	0.038	0.099
7	Rock	0.032	0.096	0.132	0.141	0.167	0.171
8	Rock	0.052	0.055	0.084	0.032	0.116	0.092
9	Rock	0.022	0.037	0.048	0.048	0.065	0.059
10	Rock	0.016	0.017	0.035	0.007	0.019	0.067
11	Rock	0.004	0.006	0.015	0.034	0.031	0.028
12	Rock	0.012	0.031	0.017	0.031	0.036	0.010
13	Rock	0.008	0.009	0.005	0.025	0.034	0.055

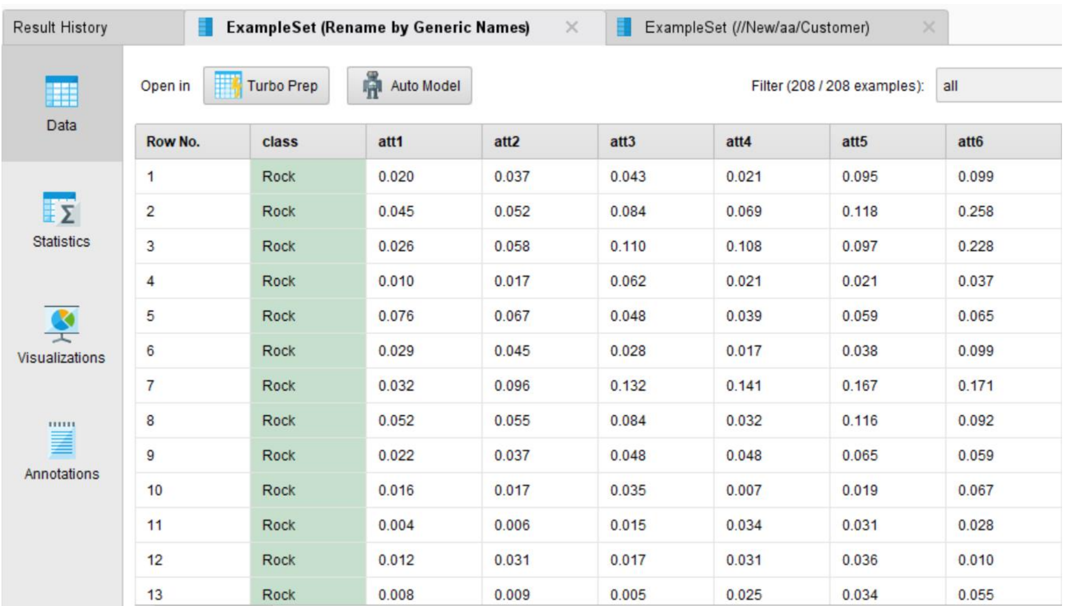
Figure 10: Result set of Rename by Replacing

The Retrieve operator is used to load the 'Sonar' data set. A breakpoint has been placed here to allow you to see the Example Set. The Example Set has 60 standard attributes with names such as attribute 1, attribute 2, and so on. On it, the Rename by Replacing operator is used. Since the attribute filter form parameter is set to 'all,' this operator can rename any attribute. The first capturing category and a slash, i.e. 'att-', are used in place of 'attribute_' in the names of the 'Sonar' attributes. As a result, attributes are renamed to att-1, att-2, and so on.

Create your dataset and rename the attributes of your dataset by considering different rename operators.

Rename by Generic Names

The Rename by Generic Names operator renames the attributes of a given ExampleSet to a collection of generic names such as att1, att2, att3, and so on. The generic name stem parameter determines the name stem to be used when constructing generic names. Using the stem 'att' as an example, attribute names will be 'att1', 'att2', and so on.



Row No.	class	att1	att2	att3	att4	att5	att6
1	Rock	0.020	0.037	0.043	0.021	0.095	0.099
2	Rock	0.045	0.052	0.084	0.069	0.118	0.258
3	Rock	0.026	0.058	0.110	0.108	0.097	0.228
4	Rock	0.010	0.017	0.062	0.021	0.021	0.037
5	Rock	0.076	0.067	0.048	0.039	0.059	0.065
6	Rock	0.029	0.045	0.028	0.017	0.038	0.099
7	Rock	0.032	0.096	0.132	0.141	0.167	0.171
8	Rock	0.052	0.055	0.084	0.032	0.116	0.092
9	Rock	0.022	0.037	0.048	0.048	0.065	0.059
10	Rock	0.016	0.017	0.035	0.007	0.019	0.067
11	Rock	0.004	0.006	0.015	0.034	0.031	0.028
12	Rock	0.012	0.031	0.017	0.031	0.036	0.010
13	Rock	0.008	0.009	0.005	0.025	0.034	0.055

Figure 11: Result set of Rename by Generic names

Rename By Example Values

The Rename by Example Values operator creates new attribute names based on the values of the listed Example Set example. Please note that all attributes, both standard and unique, have been renamed. The example is also removed from the Example Set. When an example contains the names of the attributes, this operator may be useful.

Row No.	new_label	new_name1	new_name2
1	negative	value2	value2
2	negative	value1	value4
3	negative	value4	value3
4	negative	value2	value4
5	positive	value3	value0
6	negative	value1	value2
7	negative	value1	value1
8	negative	value1	value1
9	negative	value1	value3
10	negative	value3	value4
11	positive	value0	value0
12	negative	value3	value4
13	positive	value0	value4

Figure 12: Rename By Example Values

The Sub process operator is the first step in this Example Process. The Example Set is returned by the Sub process operator. The names of the attributes are actually 'label,''att1', and'att2', as you can see. The values 'new label,'' new name1', and new name2' is used in the first example. On this Example Set, the Rename by Example Values operator is used to rename the first example's values to attribute names. The row number parameter is set to 1 in this example. You'll notice that the attributes have been renamed appropriately after the process has been completed. Furthermore, the first example from the Example Set has been deleted.

Describe different renaming operators.

8.3 Identification and Removal of Missing values in Data set

Missing values can be replaced with the Attribute's minimum, maximum, or average value. Zero may also be used to fill in gaps in data. As a substitute for missing values, any replenishment value can be listed.

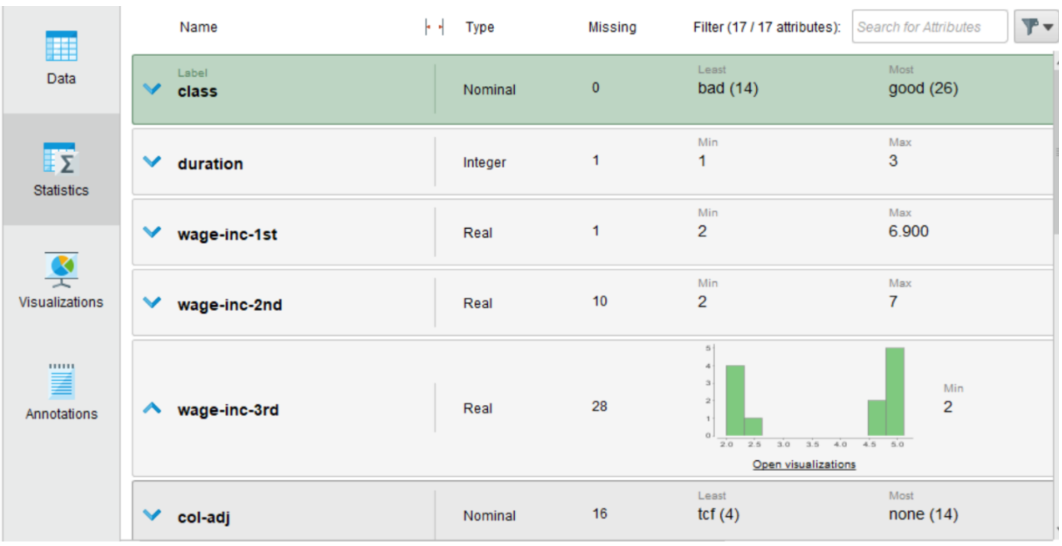


Figure 13: Dataset with Missing values

The following operators are used to handle missing values:

Replace Missing Values

This Operator replaces missing values in Examples of selected Attributes by a specified replacement.

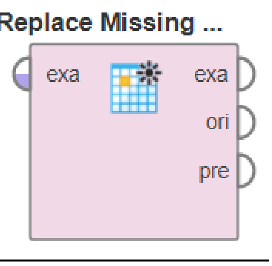


Figure 14: Replace Missing Value Operator

Missing values can be replaced with the Attribute's minimum, maximum, or average value. Zero may also be used to fill in gaps in data. As a substitute for missing values, any replenishment value can be listed.

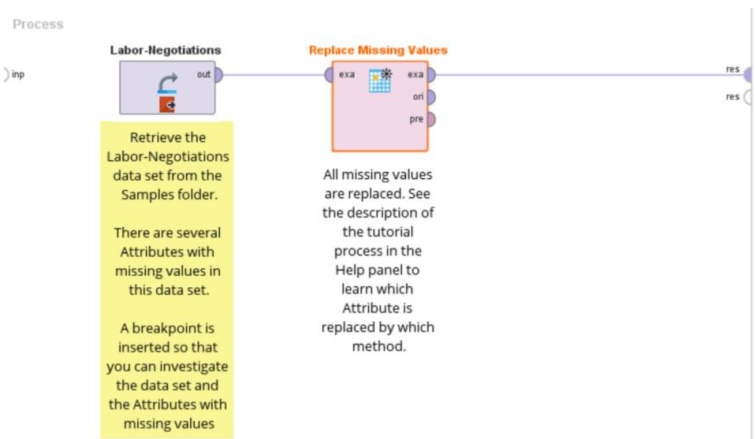


Figure 15: Design view of Replace Missing Values

This Process shows the usage of the Replace Missing Values Operator on the Labor-Negotiations data set from the Samples folder. The Operator is configured that it applies the replacement on all Attributes which have at least one missing value (*attribute filter type* is *no_missing_values* and *invert selection* is true). In the columns parameter several Attributes are set to different replacement methods:

- wage-inc-1st: minimum
- wage-inc-2nd: maximum
- wage-inc-3rd: zero
- working-hours: value

The parameter replenishment value is set to 35 so that all missing values of the Attribute working-hours are replaced by 35. The missing values of the remaining Attributes are replaced by the average of the Attribute (parameter *default*).

Name	Type	Missing	Filter (17 / 17 attributes):	Search for Attributes
Label class	Nominal	0	Least bad (14)	Most good (26)
duration	Integer	0	Min 1	Max 3
wage-inc-1st	Real	0	Min 2	Max 6.900
wage-inc-2nd	Real	0	Min 2	Max 7
wage-inc-3rd	Real	0	Min 0	Max 5.100
col-adj	Nominal	0	Least tcf (4)	Most none (30)
working-hours	Integer	0	Min 27	Max 40

Figure 16: Result after Implementing Replace Missing Values

Declare Missing Value

Declare Missing Value replaces the listed values of the selected attributes with Double.NaN, resulting in these values being marked as missing. The subsequent operators will treat these values as missing values. Nominal, numeric, and regular expression modes may be used to choose the desired values. The mode parameter can be used to manage this action.

Input

An Example Set is expected at this input port. In the attached Example Method, it's the result of the Retrieve operator. Other operators' output can also be used as data.

Output

The missing values are substituted for the required values of the selected attributes, and the resultant Example Set is provided via this port.

Original

This port passes the Example Set that was provided as input to the output without any changes. This is commonly used to reuse the same Example Set through several operators or to display the Example Set in the Results Workspace.

Parameters

attribute filter type

This parameter lets you choose the attribute selection filter, which is the tool you'll use to pick the appropriate attributes. It comes with the following features:

- **All:** This choice simply selects all of the Example Set's attributes. This is the standard-setting.
- **single**: Selecting a single attribute is possible with this choice. When you select this choice, a new parameter (attribute) appears in the Parameters panel.
- **subset:** This choice allows you to choose from a list of multiple attributes. The list contains all of the Example Set's attributes; appropriate attributes can be easily selected. If the metadata is unknown, this option will not work. When you select this choice, a new parameter appears in the Parameters panel.
- **numeric value filter**: When the numeric value filter choice is selected, a new parameter (numeric condition) appears in the Parameters panel. The examples of all numeric attributes that satisfy the numeric condition are chosen. Please note that regardless of the numerical situation, all nominal attributes are chosen.
- **no_missing_values:** This choice simply selects all of the Example Set's attributes that do not have a missing value in any of the examples. All attributes with a single missing value are deleted.
- **value_type:** This option allows you to pick all of a type's attributes. It's worth noting that styles are arranged in a hierarchy. The numeric form, for example, includes both real and integer types. When selecting attributes via this option, users should have a basic understanding of type hierarchy. Other parameters in the Parameters panel become visible when it is picked.
- **block_type:** The block type option works similarly to the value type option. This choice allows you to choose all of the attributes for a specific block form. Other parameters (block type, use block type exception) become available in the Parameters panel when this choice is selected.

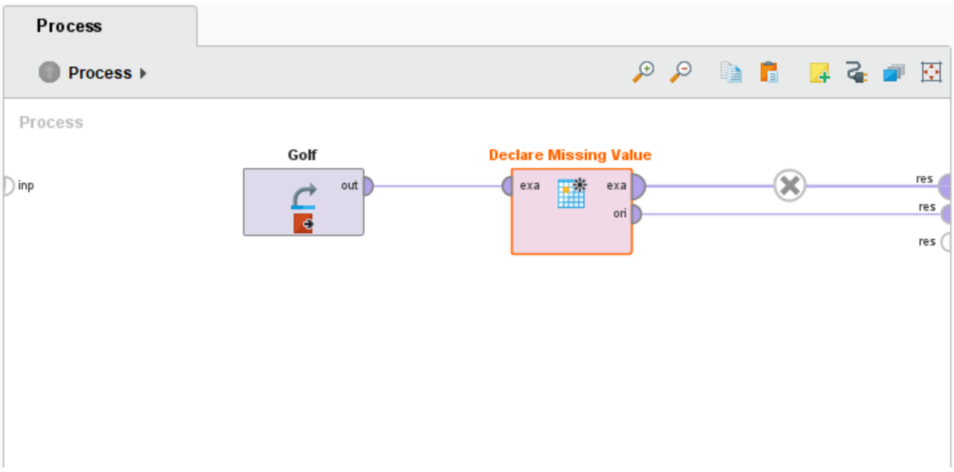


Figure 17: Implementation of Declare Missing value

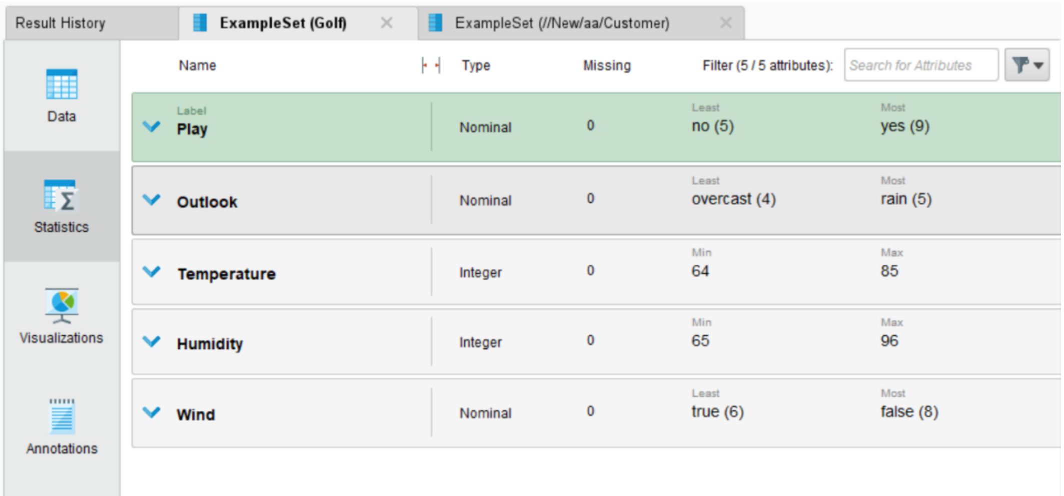


Figure 18: Result view of Declare Missing value

Replace all Missing

A universal missing value handler that replaces missing values with new ones when dealing with nominal values. MISSING uses the average of the non-missings to substitute missings and infinite in numerical columns. If a column is absent entirely, all values will be set to zero. If all dates are missing, the first known date is used, or a zero date (1 January 1970) is used again.

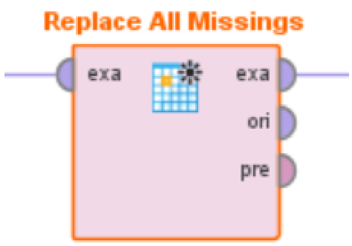


Figure 19: Replace All Missings Operator

This operator handles all missing values in a data set automatically.

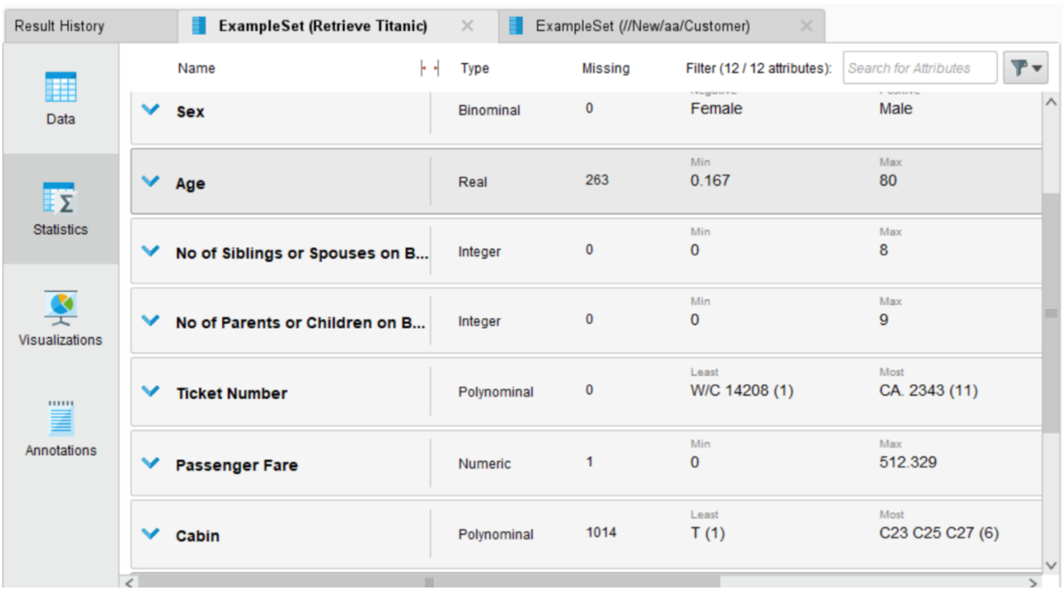


Figure 20: Before All Replace Missing

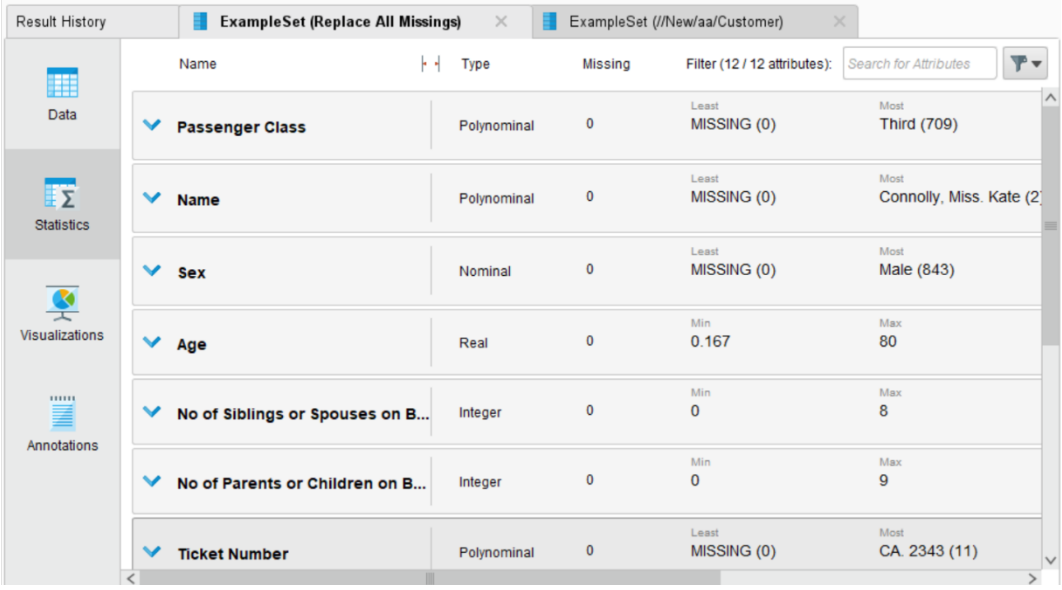
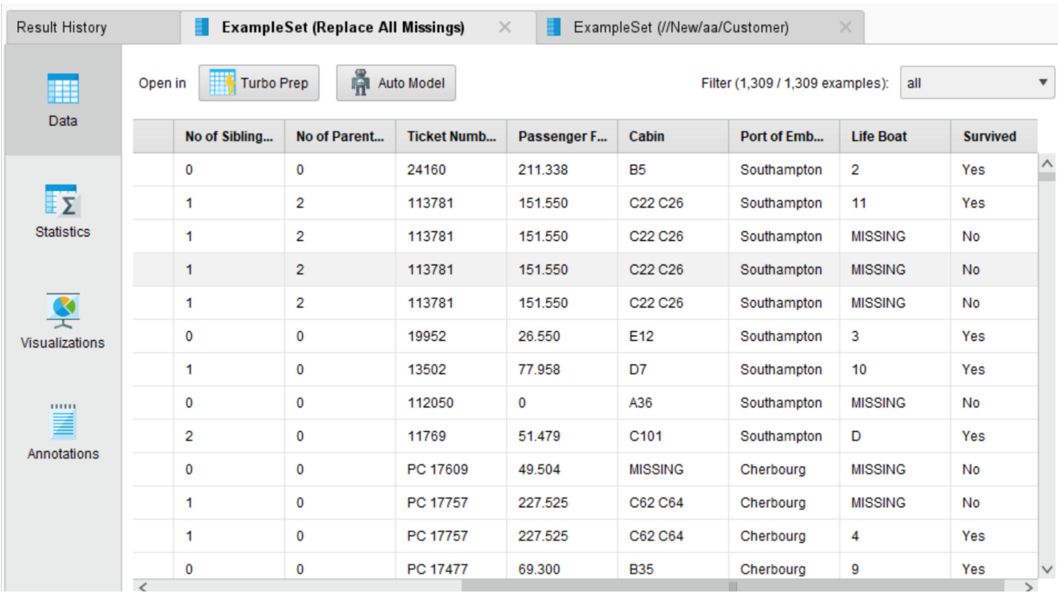


Figure 21: Result view of Relace All Missing

In the Titanic data collection, this method removes all missing values. For the remaining rows, all numerical values are replaced by average values, such as using the age 29.881 for missing ages. The term MISSING has been used to replace all nominal missings. The Titanic data set lacks dates, but they would have been replaced by the set's first date.



No of Sibling...	No of Parent...	Ticket Numb...	Passenger F...	Cabin	Port of Emb...	Life Boat	Survived
0	0	24160	211.338	B5	Southampton	2	Yes
1	2	113781	151.550	C22 C26	Southampton	11	Yes
1	2	113781	151.550	C22 C26	Southampton	MISSING	No
1	2	113781	151.550	C22 C26	Southampton	MISSING	No
1	2	113781	151.550	C22 C26	Southampton	MISSING	No
0	0	19952	26.550	E12	Southampton	3	Yes
1	0	13502	77.958	D7	Southampton	10	Yes
0	0	112050	0	A36	Southampton	MISSING	No
2	0	11769	51.479	C101	Southampton	D	Yes
0	0	PC 17609	49.504	MISSING	Cherbourg	MISSING	No
1	0	PC 17757	227.525	C62 C64	Cherbourg	MISSING	No
1	0	PC 17757	227.525	C62 C64	Cherbourg	4	Yes
0	0	PC 17477	69.300	B35	Cherbourg	9	Yes

Figure 22: Replacement of Nominal and Numerical Values

8.4 Apriori method for finding frequent item set Weka

The Apriori algorithm is a machine learning algorithm that finds likely associations and generates association rules. WEKA is a tool that implements the Apriori algorithm. While computing these guidelines, you should specify a minimum level of support and an appropriate level of confidence. The Apriori algorithm will be applied to the supermarket data given in the WEKA installation.

Loading Data

Open the Preprocess tab in the WEKA explorer, pick the supermarket.arff database from the installation folder, and press the Open file... icon. After the data has been loaded, you can see the screen below.

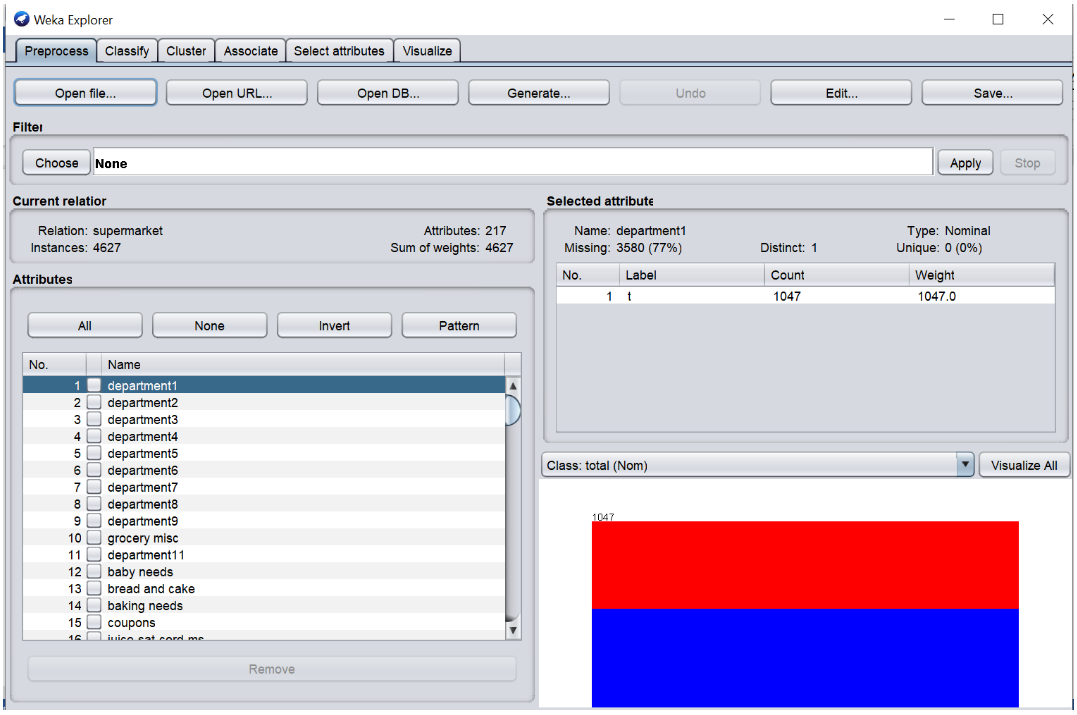


Figure 23: Loading of data

A total of 4627 instances and 217 attributes are stored in the database. It's easy to see how complicated it will be to find a connection between so many variables. The Apriori algorithm, fortunately, automates this task.

Associate

Then, on the Associate TAB, select the Choose option. As shown in the screenshot, choose the Apriori association.

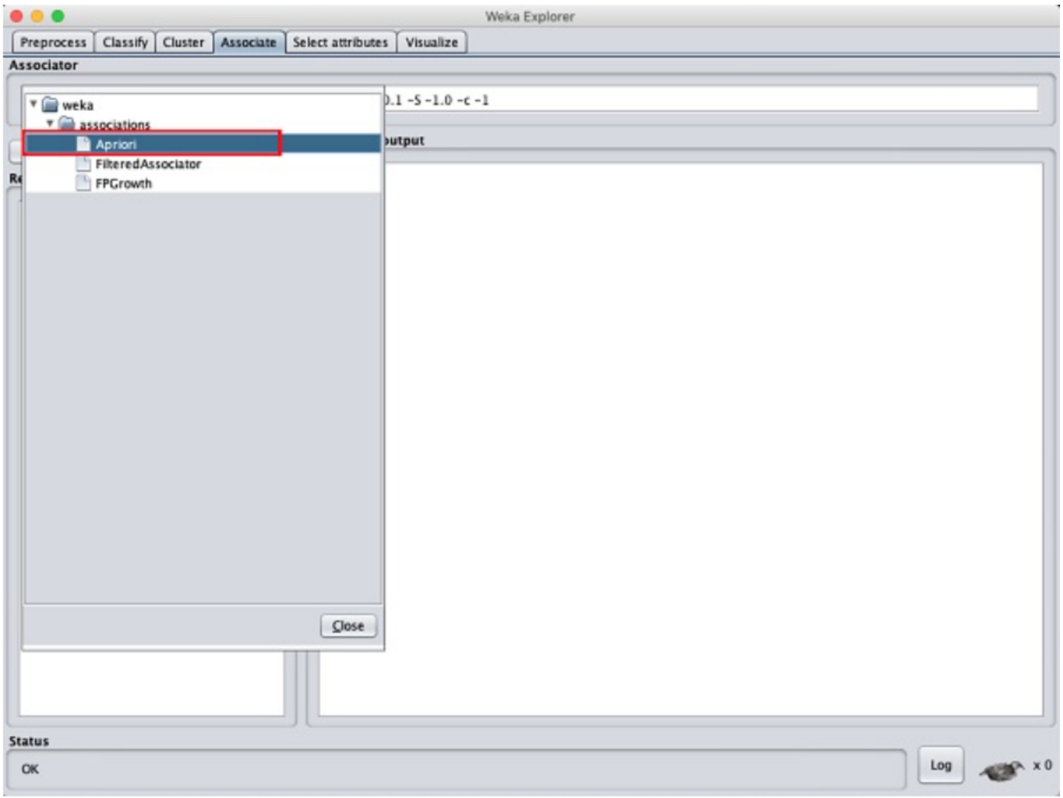


Figure 24: Selection of Apriori

To set the parameters for the Apriori algorithm, click on its name, and a window will appear, as shown below, allowing you to do so.

On the weather info, run Apriori. What is the support for this item set based on the output? outlook = rainy humidity = normal windy = FALSE play = yes.

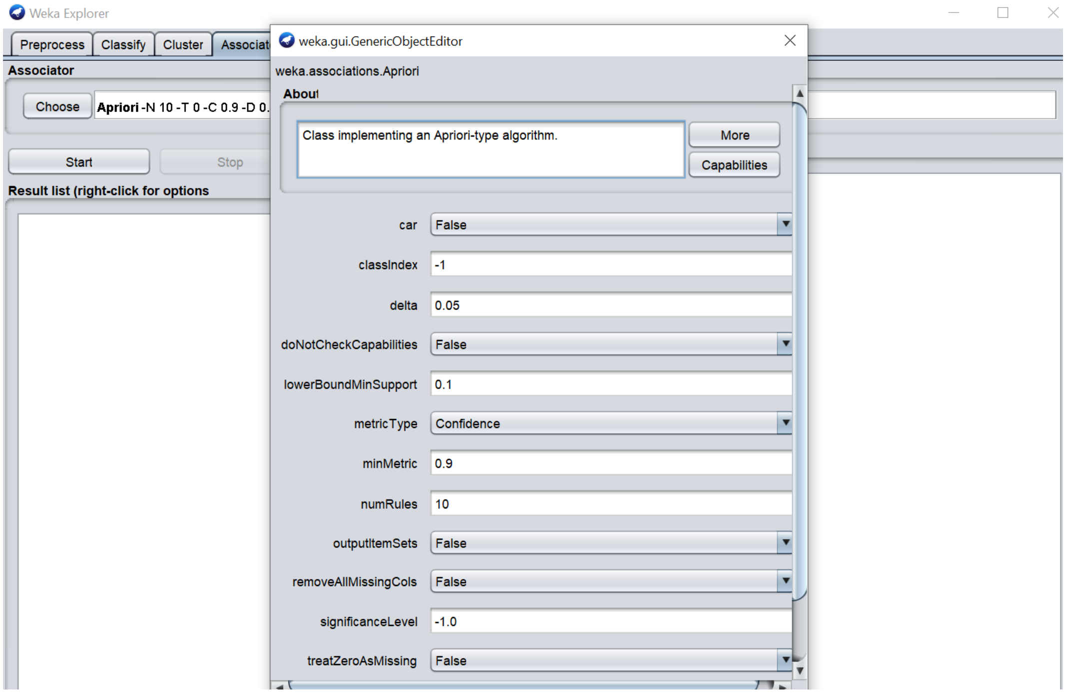


Figure 25: Parameter Setting

Click the Start button after you've set the parameters. After a while, the results will appear as shown in the screenshot below.

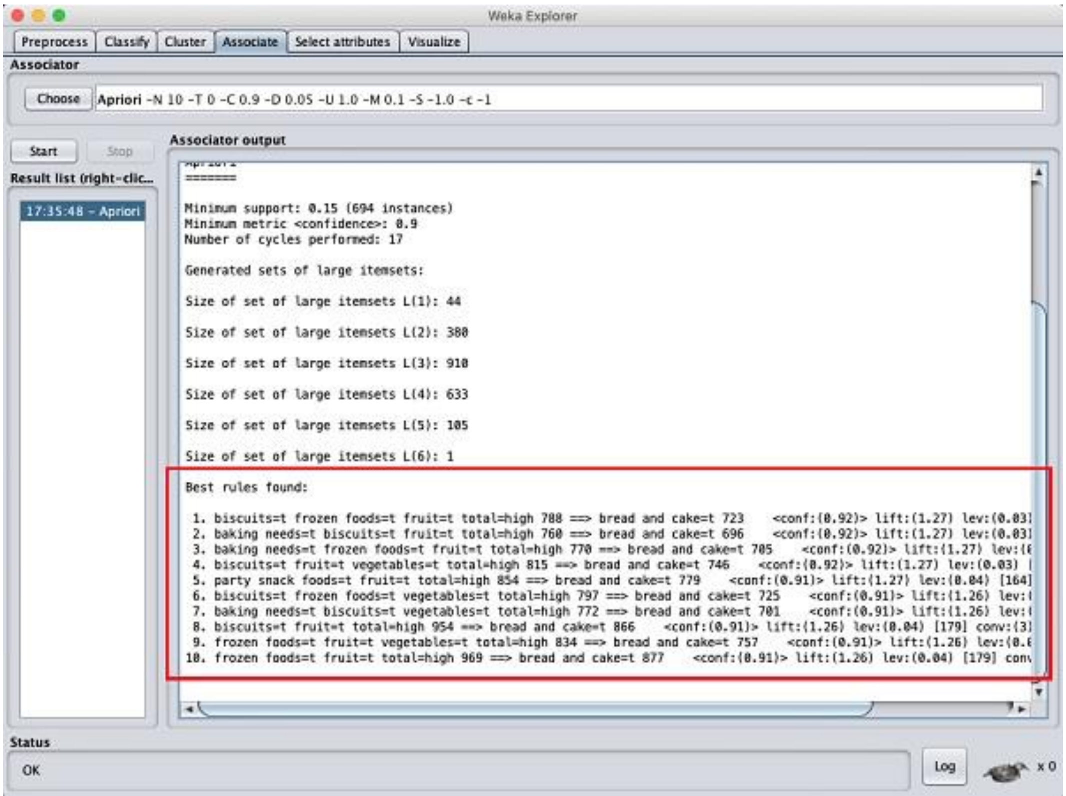


Figure 26: Apriori Results

The detected best rules of associations can be found at the bottom of the page. This will assist the supermarket in stocking the necessary shelves with their items.

Run Apriori on this data with default settings. Comment on the rules that are generated. Several of them are quite similar. How are their support and confidence values related?

Summary

- The Remove Duplicates operator compares all examples in an Example Set against each other using the listed attributes to remove duplicates.
- The special attributes are attributes with special roles which identify the examples. Special attributes are id, label, prediction, cluster, weight, and batch.
- The Rename operator is used for renaming one or more attributes of the input Example Set.
- To change the role of an operator, use the Set Role operator.
- Whitespaces, brackets, and other unnecessary characters are commonly removed from attribute names using rename by replacing.
- Missing values can be replaced with the attribute's minimum, maximum, or average value.
- Using invert selection all of the previously selected attributes have been unselected, and previously unselected attributes have been selected.

Keywords

Output Ports: The duplicate examples are removed from the given Example Set and the resultant Example Set is delivered through this port.

old name: This parameter is used to specify which attribute's name should be modified.

Rename by replacing: The Rename by Replacing operator substitutes the required replacement for parts of the attribute names.

replace what: The replace what parameter specifies which part of the attribute name should be changed.

no missing values : This choice simply selects all of the Example Set's attributes that do not have a missing value in any of the examples.

Create view : Instead of modifying the underlying data, you can build a View. To allow this choice, simply select this parameter.

Invert selection : When set to true, this parameter acts as a NOT gate, reversing the selection.

Self Assessment Questions

1. The……………operator removes duplicate examples from an Example Set by comparing all examples with each other based on the specified attributes.

A. Remove Duplicates

B. Duplicate Removal

C. Delete Duplicate

D. Erase Duplicate

2. ____________ is the default option used to remove duplicate values.

A. Subset

B. Single

C. All

D. Block_type

3. ___________ allows selection of multiple attributes through a list for the removal of duplicate values.

A. Subset

B. Single

C. All

D. Block_type

4. The ____________ operator has no impact on the type or role of an attribute.

A. Retrieve

B. Store

C. Rename

D. Filter

5. To change the role of an operator, use the ___________operator.

A. Get Role

B. Set Role

C. Change Role

D. Update Role

6. The ___________operator replaces missing values in Examples of selected Attributes by a specified replacement.

A. Replace Missing Values

B. Impute Missing Values

C. Remove Missing values

D. Hide Missing Values

7. The ____________operator estimates values for the missing values by applying a model learned for missing values.

A. Replace Missing Values

B. Impute Missing Values

C. Remove Missing values

D. Hide Missing Values

8. Which of the following attribute_filter_type allows the selection of multiple attributes through a list

A. value_type

B. subset

C. All

D. block_type

9. Which of the following attribute_filter_type option selects all Attributes of the ExampleSet which do not contain a missing value in any Example.

A. numeric_value_filter

B. regular_expression

C. no_missing_values

D. None

10. Missing data can_________ the effectiveness of classification models in terms of accuracy and bias.

A. Reduce

B. Increase

C. Maintain

D. Eliminate

11. _____________is a rule-based machine learning method for discovering interesting relations between variables in large databases.

A. Association rule learning

B. Market Based Analysis

C. Rule Learning

D. None

12. Let X and Y are the data items where ________________specifies the probability that a transaction contains X ∪ Y.

A. Support

B. Confidence

C. Both

D. None

13. Let X and Y are the data items where ____________ specifies the conditional probability that a transaction having X also contains *Y*.

A. Support

B. Confidence

C. Both

D. None

14. The……………algorithm is one such algorithm in ML that finds out the probable associations and creates association rules.

A. Decision Tree

B. KNN

C. Apriori

D. All of the above

15. You can select Apriori in Weka by clicking which of the following tab:

A. Classify

B. Cluster

C. Associate

D. Select Attribute

Answer for Self Assessment

1.	A	2.	C	3.	A	4.	C	5.	B
6.	A	7.	B	8.	B	9.	C	10.	A
11.	A	12.	A	13.	B	14.	C	15.	C

Review Questions

Q1) Why is it necessary to remove duplicate values from the dataset. Explain the step-by-step process of removing duplicate values from the dataset.

Q2) With example elucidate the different renaming operators available in rapidminer.

Q3) Explain the step-by-step process of generating association rules in weka using the apriori algorithm.

Q4) "Missing values lead to incorrect analysis of the data" Justify the statement with an appropriate example.

Q5) Elucidate the different methods of handling missing data in rapid miner.

Further Readings

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